

AMAN VERMA

Full Stack Engineer | *React.js* • *Node.js* • *NestJS* • *PostgreSQL*

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PROFESSIONAL SUMMARY

Results-driven Full Stack Engineer with 2+ years of experience architecting and delivering scalable, high-performance web applications. Proven expertise in end-to-end product development using React.js, Node.js, NestJS, and PostgreSQL, with a strong foundation in RESTful API design, real-time systems (Socket.io), and asynchronous job processing (BullMQ/Redis). Experienced in integrating AI/ML capabilities (GPT-4, Whisper) into production systems. Passionate about system design, performance optimization, and building developer-friendly backend architectures.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, HTML5, CSS3
Frontend: React.js, Redux, Bootstrap
Backend: Node.js, Express.js, NestJS, REST APIs, Socket.io, BullMQ
Databases: PostgreSQL, MySQL, Redis (caching & queues)
ORM / Tools: Sequelize ORM, Postman, Git, JWT, Role-Based Access Control
AI / Cloud: OpenAI GPT-4, Whisper (speech-to-text), Python integration

WORK EXPERIENCE

Software Engineer II

Jun 2023 – Present

Prakash Software Solutions Pvt. Ltd. — Indore, MP, India

- Architected and delivered full-stack features across 4+ production applications, owning API design, database schema, and React.js frontend integration using TypeScript.
- Engineered RESTful APIs in Node.js/Express.js and NestJS, reducing average API response time by ~30% through query optimization and Redis caching strategies.
- Designed scalable asynchronous video-processing pipelines using BullMQ queues and WebSocket-based progress updates, enabling real-time status tracking for high-volume uploads.
- Implemented JWT-based authentication and role-based access control (RBAC) across multiple modules, improving system security and reducing unauthorized access incidents to zero.
- Integrated AI capabilities (GPT-4, Whisper) into chat and video-analysis platforms, enabling real-time multilingual communication and automated computer vision reporting.
- Optimized PostgreSQL and MySQL schemas for high-volume data workloads, improving query performance by up to 40% through indexing, normalization, and caching.

PROJECTS

Video Quality Service | *React.js* • *NestJS* • *BullMQ* • *Redis* • *WebSocket* • *PostgreSQL* • *Python*

- Architected a full-stack computer vision platform enabling technicians to upload car videos for AI-driven tire gauge analysis, scratch detection, and license plate recognition.
- Built async video-processing pipelines with BullMQ, cutting end-to-end processing latency by ~50% compared to synchronous approaches; integrated WebSocket for live progress updates.
- Implemented Redis caching for high-volume API endpoints, reducing DB load by ~35% during peak upload periods.
- Delivered dashboard analytics with histogram visualizations and AI-generated summary reports including scoring metrics and detection percentages, used by 100+ dealer users.

- Enforced JWT authentication and multi-role RBAC (Admin, Dealer, Technician), securing sensitive vehicle inspection data across the platform.

Talk to Jorge | *React.js · NestJS · GPT-4 · Whisper · Socket.io · PostgreSQL*

- Developed a real-time multilingual chat platform supporting seamless text and audio message translation across 10+ languages using OpenAI GPT-4 and Whisper models.
- Integrated GPT-4 for text translation and text-to-speech, and Whisper for speech-to-text, achieving 95%+ speech recognition accuracy in user testing.
- Leveraged Socket.io for sub-100ms real-time message delivery, ensuring a smooth, lag-free communication experience for concurrent users.

Meeting Buddy | *React.js · NestJS · Socket.io · PostgreSQL*

- Built an end-to-end meeting management platform with real-time Q&A, live polls, and interactive quizzes, improving participant engagement for 500+ college students.
- Designed WebSocket-driven real-time communication layer supporting concurrent guest and attendee sessions with zero message loss under load testing.
- Implemented automated marksheet generation post-quiz with instant downloadable results, reducing manual grading effort for instructors by ~80%.

Med-Reminder | *Node.js · Express.js · MySQL · Flutter (mobile)*

- Engineered the full backend for a medication reminder mobile app, designing RESTful APIs in Node.js/Express.js for medication management, dosage scheduling, and automated reminder workflows.
- Optimized MySQL schema for efficient retrieval of user schedules and reminder history, achieving <50ms average query response time under concurrent load.
- Integrated backend seamlessly with a Flutter mobile app, ensuring real-time data sync and reliable push notifications for critical medication alerts.

EDUCATION

Master of Computer Applications (MCA)

Medi-Caps University, Indore, MP, India

Jul 2021 – May 2023